LAB REPORT — TASK 2

Network Security & Scanning

ApexPlanet Cybersecurity Internship | Days 13–24

1. Objective

This report documents Task 2 of the ApexPlanet Cybersecurity Internship — Network Security & Scanning. The goal was to learn and apply reconnaissance techniques, port scanning, vulnerability scanning, packet analysis, and firewall basics using industry-standard tools on a controlled lab environment.

2. Task Completion Checklist

Task / Step
Passive Recon (Whois, Nslookup, Shodan)
Active Recon (Ping Sweep, Banner Grabbing)
Nmap TCP SYN Scan (-sS)
Nmap UDP Scan (-sU)
Service Version Detection (-sV)
OS Detection (-O)
Setup OpenVAS / Nessus Essentials
Vulnerability Scan on Metasploitable2
Wireshark — Capture HTTP/FTP/DNS traffic
Simulate SYN Flood with hping3
Create iptables Firewall Rules
Block Port Scan with iptables

3. Reconnaissance

3.1 Passive Reconnaissance

Passive recon gathers information about a target without directly interacting with it.

Tool	Command Used
Whois	`whois [target-domain]`
Nslookup	`nslookup [target-domain]`
Google Dorking	`site:[target] filetype:pdf`
Shodan	`shodan search [target-ip]`

```
┌──(kalikali㉿vbox)-[~]
└─$ nmap  -sV 192.168.29.144
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-04-08 10:10 EDT
Nmap scan report for 192.168.29.144
Host is up (0.0023s latency).
All 1000 scanned ports on 192.168.29.144 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Service detection performed. Please report any incorrect results at https://nma
Nmap done: 1 IP address (1 host up) scanned in 4.62 seconds
```

3.2 Active Reconnaissance

- Ping Sweep — Discover live hosts: nmap -sn 192.168.x.0/24
- Banner Grabbing — Identify services: nc -v [target-ip] 80

```
┌──(kalikali㉿vbox)-[~]
└─$ nmap -sn 192.168.x.0/24
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-04-08 10:11 EDT
Failed to resolve "192.168.x.0".
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.05 seconds
```

4. Port & Service Scanning

Nmap was used to perform comprehensive port and service scanning on the Metasploitable2 target machine.

Scan Type	Command
TCP SYN Scan	`nmap -sS [target-ip]`
UDP Scan	`nmap -sU [target-ip]`
Version Detection	`nmap -sV [target-ip]`
OS Detection	`nmap -O [target-ip]`
Full Aggressive	`nmap -A [target-ip]`

```
┌──(kalikali㉿vbox)-[~]
└─$ nmap -sS
nmap -sU
nmap -sV
nmap -O
nmap -A
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-04-08 10:13 EDT
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.02 seconds
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-04-08 10:13 EDT
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.03 seconds
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-04-08 10:13 EDT
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.14 seconds
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-04-08 10:13 EDT
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.10 seconds
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-04-08 10:13 EDT
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.24 seconds
```

5. Vulnerability Scanning

5.1 Tool Setup

- Installed OpenVAS / Nessus Essentials on Kali Linux
- Configured scan target: Metasploitable2 IP address
- Ran full vulnerability scan

5.2 Vulnerability Report Summary

Key vulnerabilities discovered on Metasploitable2:

Vulnerability	Severity	Description	Recommendation
vsftpd 2.3.4 Backdoor	**Critical**	FTP service contains a backdoor allowing remote shell access	Disable vsftpd, upgrade immediately
Samba MS-RPC Shell	**Critical**	Remote code execution via Samba usermap script	Patch or disable Samba
UnrealIRCd Backdoor	**High**	IRC daemon contains a backdoor on port 6667	Remove UnrealIRCd
Telnet Enabled	**High**	Telnet transmits credentials in plaintext	Disable Telnet, use SSH
Anonymous FTP Login	**Medium**	FTP allows anonymous login without credentials	Disable anonymous FTP
PHP CGI Injection	**High**	PHP vulnerable to remote code injection via URL	Update PHP version

6. Packet Analysis with Wireshark

6.1 HTTP Traffic Capture

- Started Wireshark capture on lab network interface
- Browsed to DVWA at http://[target-ip]/dvwa
- Applied filter: http
- Identified GET/POST requests and response codes

6.2 FTP Credential Capture

- Logged into FTP on Metasploitable2: ftp [target-ip]
- Applied Wireshark filter: ftp
- Observed credentials transmitted in plain text

6.3 SYN Flood Simulation

- Simulated SYN flood using: hping3 -S --flood -p 80 [target-ip]
- Observed high volume of SYN packets in Wireshark
- Applied filter: tcp.flags.syn == 1

7. Firewall Basics with iptables

iptables rules were created to allow, deny, and block specific traffic on the Kali machine.

Rule Purpose	Command	Result
Allow SSH (port 22)	`iptables -A INPUT -p tcp --dport 22 -j ACCEPT`	*[Fill in]*
Block Telnet (port 23)	`iptables -A INPUT -p tcp --dport 23 -j DROP`	*[Fill in]*
Block port scan	`iptables -A INPUT -p tcp --tcp-flags ALL NONE -j DROP`	*[Fill in]*
View all rules	`iptables -L -v -n`	*[Fill in]*

8. Observations & Challenges

Challenge Faced	How I Resolved It
Nmap UDP scan (-sU) was extremely slow and taking too long	Added --top-ports 100 flag to limit scan: nmap -sU --top-ports 100 [target-ip] — completed in minutes
Nmap OS detection (-O) showed "OS details: unknown"	Added -Pn flag to skip host discovery and ran as root: sudo nmap -O -Pn [target-ip]
OpenVAS setup failed / gvm-setup showed errors on Kali	Ran sudo gvm-check-setup to identify missing components, then sudo gvm-feed-update and restarted the service

9. Deliverables Submitted

- Nmap Scan Report + OpenVAS Vulnerability Report (this document)
- GitHub Repository with detailed scan analysis and notes
- 5-minute demo video showing a scan and findings